AAAI-88: Seventh National Conference on Artificial Intelligence

Sponsored by the Association for the Advancement of Artificial Intelligence

The Seventh National Conference on Artificial Intelligence (AAAI-88) was held August 21–26, 1988, in Saint Paul, Minnesota.

Over the past few years, AAAI experimented with separate conference sessions for science and engineering contributions as a means of structuring presentations. Based on feedback from conference participants, we came to the conclusion that this distinction is just one of many possible dichotomies for characterizing work in our field. Therefore, this year's conference was once again structured solely around the technical subareas (e.g., computer vision, knowledge representation) that constitute the primary distinctions among the immediate goals of researchers and practitioners.

We are at an important point in the history of our discipline. Over the past few years, we have witnessed a rapid expansion of commercial interest and the application of techniques developed through the preceding decade of research. This commercial interest has significantly broadened our perspective and has deepened our understanding of methods that had previously been tested only in the laboratory. It has resulted in extensions to these techniques. It has also heightened awareness of the importance of the many basic research issues that have yet to be addressed. Whatever the cause, the field now seems to be embarking on a new examination of fundamental issues of intelligence research, which will provide a needed balance to advances in the state of AI practice.

A further significant trend that we have seen is the appearance of many small, informal workshops in specialized areas of AI. This represents an extremely healthy response to the growing size and diversity of the field. Such workshops provide opportunities for high-bandwidth, informal exchange of new ideas and conjectures in rapidly developing subareas. As such, they complement the presentation of more mature research results in the proceedings and conference sessions. We are pleased to see that the number of such workshops held in conjunction with the conference has grown from 7 in 1986 to 20 this year. These workshops added significantly to the spirit of technical exchange among all conference attendees.

Of the 850 submissions to the conference this year, we accepted 148. Each paper was reviewed by at least two members of the program committee. Our goal is to present the best: well-written papers that advance the state of the art, either theoretical or experimental.

Artificial Intelligence is a vital discipline with new directions, results, ideas, and people. We believe this conference captures the energy and intellectual excitement of the individuals working in the field.

Reid G. Smith and Tom M. Mitchell

The 1988 Best Paper Awards

The AAAI Best Paper Awards, established in 1982, recognize papers that report important, substantial research in an exemplary manner. This year 148 papers were selected by the Program Committee during the normal conference review process. Of these papers, eleven were nominated for the 1988 Best Paper Award. Soon after the Program Committee Meeting, several members of the Committee were asked to read all of the nominated papers and to indicate which of the nominated papers they felt deserved the 1988 Best Paper Award. As a result, two clear winners were identified.

This year the Program Committee Is pleased to award the Best Paper Award to the following authors:

- Steven Minton, *Carnegie Mellon University,* for "Qualitative Results Concerning the Utility of Explanation-Based Learning"
- Brian C. Williams, *Massachusetts Institute of Technology,* for "MINIMA: A Symbolic Approach to Qualitative Algebraic Reasoning"

The Best Paper Awards are sponsored by the editorial board of the journal Artificial Intelligence. In addition to providing a cash prize, Artificial Intelligence has offered each award winner the opportunity to publish a revised version of the paper in the journal. The Program Committee is also pleased to recognize the nine additional outstanding papers which were nominated for the 1988 Best Paper Award.

- Jonathan Amsterdam, *Massachusetts Institute of Technology,* for "Some Philosophical Problems with Formal Learning Theory"
- Brian Falkenhainer and Kenneth D. Forbus, *University of Illinois at Urbana Champaign,* for "Setting up Large-Scale Qualitative Models"
- John J. Grefenstette, *Naval Research Laboratory,* for "Credit Assignment in Genetic Learning Systems"

- Leslie Pack Kaelbling, *SRI International and Stanford University,* for "Goals as Parallel Program Specifications"
- Benjamin J. Kulpers and Yung-Tal Byun, *University of Texas at Austin,* for "A Robust, Qualitative Method for Robot Spatial Learning"
- Mark F. Orelup, John R. Dixon and Paul R. Cohen, *University of Massachusetts Amherst,* and Melvin K. Simmons, *General Electric Corporate Research and Development,* for "Dominic II: Meta-Level Control in Iterative Redesign"
- David B. Searls, *Unisys Paoli Research Center,* for "Representing Genetic Information with Formal Grammars"
- Reid G. Simmons, *Massachusetts Institute of Technology,* for "A Theory of Debugging Plans and Interpretations"
- L.G. Valiant, *Harvard University,* for "Functionality in Neural Nets"

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